Net Thrust from a Yunshao-Field Gradient in a Closed Cavity: A Decisive Experiment

Abstract

Cavity-thruster experiments of the EM Drive type reported a net thrust of about $1.2\ \mathrm{mN/kW}$ at NASA Eagleworks in 2016, and were driven to zero by the Tajmar group at TU Dresden in 2021 after thermal expansion, cable forces and residual gas were eliminated. Both data sets are real; the conclusions are opposite. The reason is that no published experiment has measured the one signature that separates the two: a thrust that reverses sign with the material gradient.

We propose a decisive experiment. On the mechanism side we use the gradient coupling of the Yunshao field: the topological angle θ varies in space, and its gradient multiplies $\mathbf{E}\cdot\mathbf{B}$ inside the cavity to produce a force density that can be positive or negative; a uniform θ gives zero net force. The coefficient in this expression is not a free parameter: the theta-term coefficient is the same constant as the quantized magnetoelectric response, which has already been measured in topological insulators to equal the fine-structure constant [7,8]; dividing the two reported values gives $\kappa = 1.00 \pm 0.01$ (Section 2.1). What the experiment measures is not the coefficient but whether the cavity-integrated net thrust reaches the predicted value.

The apparatus uses two cavities of identical geometry, mass and absorbed power. One carries a gradient liner, the other a uniform liner, and they are mounted back to back on a vacuum torsion balance. Three independent checks follow. First, reversing the gradient liner end for end must reverse the thrust, while thermal expansion does not reverse. Second, sweeping frequency at constant absorbed power must collapse the field response along a Lorentzian while the thermal signal stays flat. Third, changing the cavity Q at constant absorbed power must scale the thrust linearly in Q while the thermal signal stays put. If any two of the three fail, the mechanism is falsified.

For $\kappa = 1$, a 1 kW cavity with $Q = 10^4$ and $0.01\ \mathrm{m}^3$ gives about $32\ \mu\mathrm{N}$, an order of magnitude above the noise floor demonstrated by Tajmar et al. The experiment can therefore choose between "zero" and "natural coupling"; it does not add another unattributable small signal.

Keywords: propellantless propulsion; Yunshao field; theta term; axion electrodynamics; torsion balance; pre-registration; artefact control

1 The problem: a machine that failed replication

In 2001 Shawyer proposed a conical microwave cavity that would produce net thrust. In 2016 the Eagleworks group at NASA Johnson Space Center published a vacuum torsion-balance measurement reporting $1.2 \pm 0.1\ \mathrm{mN/kW}$ at 2.45 GHz and about 1 kW input [1]. The Cannae drive, tested in the same period, reported signals of tens of micronewtons.

In 2021 the Tajmar group repeated the measurement with a higher-precision torsion balance. After thermal expansion, cable Lorentz forces and residual-gas convection were removed one by one, the signal went to zero

[2]. Their conclusion was that every earlier positive result can be explained by an external force.

The two data sets do not contradict each other: the first did not push thermal effects below the signal, the second did. The question that remains unanswered is this: **if the thrust really comes from a coupling inside the cavity that depends on the material, it should reverse sign when a controllable material property is reversed; if it comes from heat, it will not.** No published experiment has performed that sign-reversal test.

This paper turns that test into an executable experiment.

2 Hypothesis: the Yunshao field and the force density

In the literature on topological insulators and axion electrodynamics, the electromagnetic action can carry an extra term [3,4]:

$$S_\theta = \int d^4x \, \frac{\theta(x)}{2\pi} \, \frac{e^2}{h} \, \mathbf{E} \cdot \mathbf{B}.$$

The Yunshao framework calls the field associated with this term the **Yunshao field**, and writes its topological coupling angle as θ. In the older literature the same object is called the axion field and θ the axion angle; both names are used when citing that literature. When θ is constant the term is a total derivative and does not affect the equations of motion; when θ has a spatial gradient it exerts a force density on the medium. Variation gives

$$\mathbf{f} = \kappa \, \frac{e^2}{2\pi h} \, \nabla\theta \, (\mathbf{E} \cdot \mathbf{B}),$$

where κ is a dimensionless coefficient. In topological materials it is not free: the theta-term coefficient is the same constant as the quantized magnetoelectric response, measured in Bi_2Se_3 and similar materials in 2016–2017 to equal the fine-structure constant [7,8], so $\kappa = 1$ is the standard strength, not an assumption to be tested. Disorder or magnetic doping in a real liner may add corrections; that is a sample-characterisation issue and does not change the coefficient at this level. Dimensional check: e^2/h is a conductance (3.874×10^{-5} S), $\nabla\theta$ is m^{-1}, and $\mathbf{E} \cdot \mathbf{B}$ is $\mathrm{V} \cdot \mathrm{T}/\mathrm{m}$; the product is N/m^3, a force density.

This expression is the standard result of axion electrodynamics. Brevik and Chaichian write the force density on the medium as $\mathbf{f} = \rho\mathbf{E} + \mathbf{J} \times \mathbf{B} - g(\mathbf{E} \cdot \mathbf{B})\nabla\theta$; substituting the medium charge and current and applying one vector identity gives the expression above [6]. κ only rescales the standard coefficient; with the measured value of the same coupling, $\kappa = 1$.

2.1 The value of κ, computed from two published magnetoelectric measurements

κ is defined as the measured theta-term coefficient divided by the standard coefficient α. Any published magnetoelectric measurement that reports a number therefore fixes it, and no new measurement is needed.

The first data set is Wu et al., 2016, on Bi_2Se_3 films [7]. Above 5 T the Faraday and Kerr rotations reach quantized plateaus; combining the two rotations to cancel the substrate index, they extract a best value of $1/137.9$ for the fine-structure constant, against the accepted $1/137.04$, a deviation the paper states as about 0.5%. By definition this gives

$$\kappa = \frac{1/137.9}{1/137.04} = 0.9937.$$

The second data set is Dziom et al., 2017, on a 58 nm HgTe film [8]. At 0.35 THz and above 5 T the Faraday rotation saturates at $|\theta_F| = \arctan\alpha \approx 7.3 \times 10^{-3}$ rad, unchanged as the gate voltage runs from 0.7 V to 1.9 V. With $\alpha = 7.2974 \times 10^{-3}$ this gives

$$\kappa = \frac{7.3 \times 10^{-3}}{7.2974 \times 10^{-3}} = 1.0004.$$

Combining the two, $\kappa = 1.00 \pm 0.01$. Propagated to the thrust, the uncertainty on $31.6\ \mu\mathrm{N}$ is about $0.3\ \mu\mathrm{N}$, the same size as the noise floor quoted in Section 3.2, so the decision is unaffected. Even with a conservative 3% allowance for thin-film corrections, the prediction becomes $31.6 \pm 0.9\ \mu\mathrm{N}$, still about 30 times the noise floor.

The same two experiments also report a plateau index one higher than the filling factor, half per surface. That offset fixes the topological value of θ at π, which is the basis for sweeping the liner from 0 to π in Section 3.1.

One external dispute should be stated. Berger et al., 2024, argue that in the thin-film limit of a 3D topological insulator an optical experiment alone cannot establish the topological magnetoelectric effect [10]. This does not affect the conversion here: whether the thin film shows the surface half-integer quantum Hall response or the bulk theta term, both come from the same action and share the same coefficient, differing only in the effective domain of θ. Once the rotation is pinned to α, the coefficient in front of the force density is pinned to 1.

Three properties matter:

1. When $\nabla\theta = 0$ the net force is zero. No matter how asymmetric the fields inside the cavity are, a uniform θ integrates to no net force. This does not conflict with momentum conservation: the force acts on the medium, and the medium exchanges momentum with the cavity.
2. The force reverses with $\nabla\theta$. Turn the liner gradient around and the force reverses; thermal expansion always points outward and does not reverse.
3. The force scales with $\mathbf{E}\cdot\mathbf{B}$, hence with cavity Q and absorbed power; a thermal effect scales only with absorbed power.

Two of these properties already have experimental anchors. Wu et al. measured quantized Faraday and Kerr rotation in Bi_2Se_3 films [7]; Dziom et al. measured a Faraday rotation angle equal to the fine-structure constant in a 3D topological insulator [8]; in Weyl semimetals the spatial gradient $\theta(\mathbf{r}) = 2\mathbf{b}\cdot\mathbf{r}$ is observed through the chiral-anomaly negative magnetoresistance [9]. The theta term and its gradient have both been measured. What has not been measured is whether they give a net thrust to a closed cavity.

The Yunshao framework states these three properties as design rules, with an internal success estimate of 5–10% at system level [5]. This paper keeps that honesty boundary: the coupling layer takes $\kappa = 1.00 \pm 0.01$, computed from published measurements (Section 2.1); the engineering layer promises no particular 32 μN result, only that Sections 3 and 4 will be executed and all data released.

3 The decisive experiment: three chains

3.1 Apparatus

Two identical cylindrical cavities, inner diameter 0.28 m, length 0.30 m, volume about 0.01 m³, operating in the TM010 mode at 2.45 GHz with an unloaded Q of about 10^4. The two cavities are mounted back to back on the same torsion rod, with opposite axial orientations so that thermal expansion and radiation pressure cancel to first order.

Cavity A: a removable gradient liner on the inner wall, with θ running from 0 to π along the axis, giving $\nabla\theta \approx \pi/L = 10.5\ \mathrm{m}^{-1}$. Cavity B: a liner of identical mass, thickness and absorptivity, but with uniform θ ($\theta = \pi/2$ everywhere), serving as the "same heat, no gradient" control. A third liner C is kept on hand: same material, mass and absorptivity as A, with uniform θ, to separate "material" from "gradient".

Each cavity is driven continuously by its own 1 kW magnetron, with both locked to a common regulated supply; the difference in absorbed power is monitored by directional couplers and held below 1%. Runs alternate in one-hour blocks: ten blocks with liner A, then the liner is turned end for end and ten more blocks are taken, with one hour of thermal settling between the two sets. The assembly sits in a 10^{-6} Pa vacuum tank with two layers of mu-metal shielding; the torsion balance is read by an optical lever at 10 Hz. Liners are numbered by a third party, and the analyst does not know which orientation a block belongs to until the data are unblinded.

3.2 Thrust chain

First measure the differential force between A and B. If the mechanism holds,

$$F_A - F_B = \kappa \, \frac{e^2}{2\pi h} \, \nabla\theta \, \langle \mathbf{E} \cdot \mathbf{B} \rangle \, V .$$

With the numbers of Section 5 this is about $32\ \mu\mathrm{N}$ for $\kappa = 1$, and zero if the mechanism does not exist. The demonstrated force noise of a torsion balance in the 0.01–1 Hz band is of order $0.3\ \mu\mathrm{N}$ [2], so the signal-to-noise ratio is about 100.

3.3 Signature chain

Three independent signatures must all hold:

Signature one: gradient reversal. Turn the liner of cavity A end for end, keeping geometry, mass and absorptivity unchanged. F_A must reverse. Thermal expansion, radiation pressure and gas flow do not reverse with the liner. This is the strongest test: it separates "material gradient" from "geometric asymmetry".

Signature two: constant-absorption frequency sweep. Sweep from 2.40 to 2.50 GHz in 0.5 MHz steps. At each point adjust the input power so that the net absorbed power stays at 1 kW. If the force comes from the Yunshao-field response it collapses along a Lorentzian with detuning; if it comes from heat, constant absorbed power means constant force. The two curve shapes are distinguishable.

Signature three: Q scaling. Change the insertion depth of the coupling probe to move Q from 5×10^3 to 2×10^4 while keeping absorbed power at 1 kW. The Yunshao-field force grows linearly with Q (because $\mathbf{E} \cdot \mathbf{B} \propto Q$); a thermal effect does not move with Q. Measure three Q points and take the slope. Radiation pressure also grows with stored energy and therefore with Q; it cancels to first order in the symmetric twin cavity, and to guard against a residual, insert a block with the uniform liner C at the same Q as a zero point.

3.4 Ledger chain

Before the first run, the following criteria are written into a pre-registration file with a notarised timestamp:

- Primary criterion: the 95% confidence interval on the measured thrust divided by the $\kappa = 1$ prediction (the "effective κ").
- Falsification line: if the effective $|\kappa| < 0.05$, the hypothesis is rejected for this material class; the authors commit to a public retraction and to stopping the line.
- Confirmation line: if the effective $\kappa > 0.5$ and all three signatures pass, a second round of independent replication is required, performed by a third party not involved in the first round.
- All data and code are released, including raw torsion waveforms, thermistor records and magnetometer records.
- Momentum ledger: if a net force is measured, strain gauges at the liner mounts must simultaneously read an equal and opposite force; if they do not, that block is discarded.

These three chains of thrust, signature and ledger mirror the three-chain criteria of the Yunshao Qiaofu document [5].

4 Artefact list and paired controls

Artefact	Magnitude	Control
Thermal expansion	~50 μm for a 0.3 m copper cavity at 1 kW and 10 K rise	Twin cavities back to back cancel to first order; wait 1 h for thermal steady state; eight thermistors recorded
Cable Lorentz force	scales as current squared, up to tens of μN	Fixed cable routing, twisted pairs, coaxial return; measure before and after rewiring
Residual gas convection	negligible below 10^{-3} Pa	Vacuum better than 10^{-6} Pa; pressure scan from 10^{-6} to 10^{-4} Pa
Radiation pressure	~6.7 μN for 1 kW fully reflected	Symmetric cavities cancel; replace with an absorbing load as a control
Magnetostriction and geomagnetism	scales as B^2	Two mu-metal layers; remanence below 1 μT
Ground vibration	large at low frequency	Torsion mode at 0.2 Hz, away from building peaks; active isolation table

The purpose of this table is not to list possibilities but to require that every artefact has a paired control that can drive its contribution to zero. If a control cannot be run, that data set does not enter the primary criterion.

5 Expected signal and decision

Take $\nabla\theta = \pi/0.3 = 10.47\ \mathrm{m^{-1}}$, $V = 0.01\ \mathrm{m^3}$, $P = 1\ \mathrm{kW}$, $Q = 10^4$, $\omega = 2\pi \times 2.45\ \mathrm{GHz}$:

Energy density

$$u = \frac{PQ}{\omega V} = 6.50 \times 10^{-2}\ \mathrm{J/m^3},$$

electric field

$$E = \sqrt{2u/\varepsilon_0} = 1.21 \times 10^{5}\ \mathrm{V/m},$$

magnetic field $B = E/c = 4.04 \times 10^{-4}\ \mathrm{T}$, hence

$$\langle \mathbf{E} \cdot \mathbf{B} \rangle = 48.9\ \mathrm{V \cdot T/m},$$

force density

$$f = 6.166 \times 10^{-6} \times 10.47 \times 48.9 = 3.16 \times 10^{-3}\ \mathrm{N/m^3},$$

total force

$$F = fV = 3.16 \times 10^{-5}\ \mathrm{N} = 31.6\ \mu\mathrm{N} \quad (\kappa = 1).$$

At $Q = 10^5$ and the same absorbed power, F rises to about 316 μN. This range falls exactly in the window where thermal artefacts can be pushed below 1 μN and a torsion balance can still resolve the signal, so the experiment has decision power.

For scale: $31.6\ \mu\mathrm{N}$ on a one-tonne payload gives $3.2 \times 10^{-8}\ \mathrm{m/s^2}$, about 1 m/s per year. It is not a usable engine; it is an instrument that can answer "whether".

6 Relation to earlier experiments

Converting the NASA 2016 value of $1.2\ \mathrm{mN/kW}$ into an effective κ at their $Q \approx 10^4$ gives about 38. With the measured coupling $\kappa = 1$, the mechanism of this paper delivers only the 32 μN scale, about 38 times below the NASA report; if the old data are real they need a stronger mechanism than this one layer of coupling. If this experiment measures an effective $\kappa \approx 1$, the old data can only be artefacts, consistent with the Tajmar attribution; only a value clearly above 1 would call for a new explanation. Either outcome is an advance over the current standoff.

To be explicit: this paper does not claim that the NASA data are correct, nor that the Tajmar attribution is wrong. It claims only that between the two explanations there exists a cheap, decisive experiment that no one has yet performed.

7 Boundaries and commitments

1. The coupling coefficient is $\kappa = 1.00 \pm 0.01$, computed from two published magnetoelectric measurements (Section 2.1). The uncertainty in this paper lies in geometric factors, material corrections and the cavity integral, not in the coupling. The 32 μN figure is a prediction for ideal geometry, not a promise.
2. A defect array with 0.125 nm period has no public fabrication or replication record. This paper treats the material as part of the experiment, not as an established fact.

3. The internal success estimate is 5–10% at system level (Yunshao framework, 2026-09-30) [5]. The number is not flattering, but it is the number the authors are willing to sign.
4. If any two of the three signatures fail, the authors commit to publishing the falsification within 30 days and to stopping the line, rather than extending it on the grounds that "more parameters are needed".
5. This paper does not seek funding. It seeks one independent measurement executed against the criteria above. Any institution that performs Sections 3 and 4 will be accepted, whatever the result.
6. This proposal does not explain where the momentum goes. If the net force is real, the momentum must be carried by the Yunshao field or by the medium; the experiment checks both a strain-gauge channel and a cavity-displacement channel, and a mismatch is scored as an artefact. This is the same rule as the momentum ledger in Section 3.4, written twice because it is the easiest one to skip.

8 Conclusion

Yunshao-field gradient coupling gives a force density that can change sign, reverses with the material, and scales with Q. The coupling layer is $\kappa = 1.00 \pm 0.01$, computed from published measurements; what remains unknown is the cavity integral and the material correction. These three properties make it experimentally separable from thermal artefacts. The differential twin cavity, gradient reversal, constant-power sweep and Q scaling described here cost on the order of one million RMB and about 18 months, and have enough decision power to close or open the route.

If the result is zero, we lose one experiment and one hypothesis. If it is not zero, we gain an instrument that can measure a new coupling. Either outcome is worth more than another argument about a 2016 machine.

Appendix A Numerical check

Quantity	Value	Source
e^2/h	$3.8740 \times 10^{-5}\ \mathrm{S}$	fundamental constants
$e^2/2\pi h$	$6.1657 \times 10^{-6}\ \mathrm{S}$	above divided by 2π
$\nabla\theta$	$10.47\ \mathrm{m^{-1}}$	$\pi/0.3\ \mathrm{m}$
u	$6.50 \times 10^{-2}\ \mathrm{J/m^3}$	$PQ/(\omega V)$
E	$1.21 \times 10^{5}\ \mathrm{V/m}$	$\sqrt{2u/\varepsilon_0}$
B	$4.04 \times 10^{-4}\ \mathrm{T}$	E/c
$\mathbf{E}\cdot\mathbf{B}$	$48.9\ \mathrm{V\cdot T/m}$	EB
f	$3.16 \times 10^{-3}\ \mathrm{N/m^3}$	$\kappa = 1$
F	$31.6\ \mu\mathrm{N}$	fV
α_{meas} (Wu 2016)	$1/137.9$	from combined Faraday and Kerr rotations; stated deviation about 0.5% from $1/137.04$

Quantity	Value	Source
κ (Wu 2016)	0.9937	$(1/137.9)/(1/137.04)$
κ (Dziom 2017)	1.0004	$7.3 \times 10^{-3}/7.2974 \times 10^{-3}$
κ adopted	1.00 ± 0.01	rounded from the two; force uncertainty about $0.3\ \mu\mathrm{N}$

Appendix B Material parameters

- Defect array spacing $d = 0.125\ \mathrm{nm}$, cross angle 45°, from the Yunshao material specification [5].
- Reference frequency $f_0 = 373\ \mathrm{THz}$; default write frequency candidate $31.08\ \mathrm{THz} = f_0/12$, to be placed inside the measured FTIR phonon band [5].
- Topological-insulator control: Bi_2Se_3 and Bi_2Te_3, with switchable $\theta = \pi$ and $\theta = 0$ states [3].

Appendix C Framework sources

The formulas, constants and honesty boundary in Sections 2 and 7 are taken from the internal Yunshao document:

- *Yunshao Gravitational Thruster Technical Document v2.0*, 2026-09-30, lines 57, 136, 206, 412 and 1539 (Yunshao-field theta term, action, thrust formula, honesty statement, Sanlian criteria).
- Same document, lines 268 and 272: $d = 0.125\ \mathrm{nm}$, $f_0 = 373\ \mathrm{THz}$.
- Same document, line 372: the gravitational-anomaly prediction was broken by internal review and reclassified as an independent new-physics bet.

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